

A weakly closed counterexample for remotality of convex hulls

arXiv automatic math run `fa_banach_001`, agent `agent_super_01`

July 1, 2026

Status

This packet records a candidate counterexample, likely valid, to a question in Martin–Rao [1]. The source paper asks whether, for a weakly closed and bounded set E , remotality of $K = \overline{\text{co}}(E)$ from a point forces remotality of E from that point. It also asks whether the answer is positive for RNP sets. The construction below gives a negative answer to both versions.

Source question

The question appears in Remark 8 of [1], immediately after an example showing that norm closedness is not enough:

Let X be a Banach space and let $E \subset X$ be a weakly closed and bounded set. We do not know if remotality of $K = \overline{\text{co}}(E)$ from a point always implies that of E . [...] We may also ask whether the above question has a positive answer for RNP sets.

The source paper’s main theorem solves the headline problem of existence of closed bounded convex non-remotal sets in every infinite-dimensional Banach space; the remark above is a separate residual question about passing remotality from a closed convex hull back to its generating set.

Definitions

A bounded set $A \subset X$ is *remotal from* $x \in X$ if there is $a_0 \in A$ such that

$$\|x - a_0\| = \sup\{\|x - a\| : a \in A\}.$$

It is enough to consider $x = 0$, since the general case is obtained by translation.

We use $K = \overline{\text{co}}(E)$ for the norm-closed convex hull. A closed bounded convex set is an RNP set if it has the Radon–Nikodym property in the standard convex set sense; in particular every closed bounded convex subset of a Banach space with the Radon–Nikodym property is an RNP set.

Proof intuition

Martin–Rao’s norm-closed example fails to be weakly closed because its Hilbert basis coordinate converges weakly to zero, and the missing weak limit is exactly the farthest point created in the convex hull. The fix is to attach a second, smaller “tag” coordinate which prevents weak convergence of the individual generators, but which can still be cancelled by taking signed midpoints.

The summing basis $s_n = (1, \dots, 1, 0, \dots)$ in c_0 has precisely this behavior: it has no weak limit in c_0 , since coordinate functionals would force any limit to be the constant sequence 1, but a signed pair $\pm s_n$ cancels exactly. Pairing this tag with the usual Hilbert basis coordinate gives a weakly closed set E ; averaging each signed pair gives $(a_n, 0, 0)$, and these averages converge in norm to a unit vector in $\overline{\text{co}}(E)$.

The counterexample

Theorem 1. *There is a Banach space X , a weakly closed bounded set $E \subset X$, and $K = \overline{\text{co}}(E)$, such that K is remotal from 0 but E is not remotal from 0. Moreover, K may be chosen to be an RNP set.*

Proof. Let

$$X = \mathbb{R} \oplus_{\infty} \ell_2 \oplus_{\infty} c_0$$

with norm

$$\|(t, h, z)\| = \max\{|t|, \|h\|_2, \|z\|_{\infty}\}.$$

Let (e_n) be the canonical orthonormal basis of ℓ_2 , and let

$$s_n = (\underbrace{1, \dots, 1}_{n \text{ times}}, 0, 0, \dots) \in c_0$$

be the summing basis of c_0 . Put $a_n = n/(n+1)$ and fix $c = 1/2$. Define

$$x_n^+ = (a_n, a_n e_n, c s_n), \quad x_n^- = (a_n, -a_n e_n, -c s_n)$$

and

$$E = \{x_n^+, x_n^- : n \in \mathbb{N}\}.$$

The set E is bounded and every element of E has norm

$$\|x_n^{\pm}\| = \max\{a_n, a_n, c\} = a_n < 1.$$

Hence E has no vector of maximal norm, while

$$\sup_{x \in E} \|x\| = \sup_n a_n = 1.$$

Thus E is not remotal from 0.

We next prove that E is weakly closed. Suppose a net $(x_{n_i}^{\sigma_i})_i$, with $\sigma_i \in \{+, -\}$, converges weakly in X to some point x . If the indices n_i are eventually contained in a finite subset of $\mathbb{N}$, then a subnet is eventually constant and the weak limit belongs to the finite set E . So it remains to rule out the case in which n_i tends to infinity along a subnet.

Passing to this subnet, the c_0 -coordinates

$$\sigma_i c s_{n_i}$$

would converge weakly in c_0 . Evaluating at the first coordinate functional $\delta_1 \in \ell_1 = c_0^*$, the scalars $\delta_1(\sigma_i c s_{n_i}) = \sigma_i c$ must converge. Since they take only the two values $\pm c$, they are eventually constant; write the eventual sign as σ . For every fixed k , once $n_i \geq k$ we have

$$\delta_k(\sigma_i c s_{n_i}) = \sigma c.$$

Therefore a weak limit $z \in c_0$ would satisfy $z_k = \sigma c$ for every k , impossible because $c > 0$ and the constant nonzero sequence is not in c_0 . Thus no such subnet exists, and E is weakly closed.

Now let $K = \overline{\text{co}}(E)$. For each n ,

$$\frac{x_n^+ + x_n^-}{2} = (a_n, 0, 0) \in \text{co}(E).$$

Since $a_n \rightarrow 1$, the point

$$u = (1, 0, 0)$$

belongs to K . Also $E \subset B_X$, so $K \subset B_X$. Hence

$$\sup_{y \in K} \|y\| = 1 = \|u\|,$$

and K is remotal from 0.

Finally, $X^* = \mathbb{R} \oplus_1 \ell_2 \oplus_1 \ell_1$ is separable. Therefore X has the Radon–Nikodym property. Since K is a closed bounded convex subset of X , it is an RNP set. This gives the asserted negative answer even for RNP sets. $\square$

Verification and novelty checks

The proof is deterministic and uses only elementary properties of ℓ_2 , c_0 , and the Radon–Nikodym property. The key point is weak closedness of E , supplied by the summing-basis tag in c_0 .

Before promotion, the following bounded novelty checks were performed. The run indexes

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`

were searched for arXiv:0909.1992, remotality, remotal sets, farthest points, closed convex hulls, weakly closed bounded sets, and RNP-set variants. No existing answer to this source question was found. The local parsed arXiv corpus was searched for remotality, farthest point, weakly closed, closed convex hull, and Martin–Rao/Sababheh–Khalil phrases; the visible later material referenced the main Martin–Rao theorem but did not answer Remark 8. External web searches used exact phrases such as “We do not know if remotality” with “`cconv(E)`”, “remotality” with “weakly closed” and “closed convex hull”, and “remotal” with “RNP” and “closed convex hull”. These searches found no later solution to the specific weakly closed/RNP convex-hull question.

Human-review recommendation

Review as a likely valid counterexample. The two points most worth checking are:

1. the net argument proving that E is weakly closed;
2. the interpretation of the source phrase “for RNP sets”. The packet proves the stronger-looking formulation where $K = \overline{\text{co}}(E)$ is a closed bounded convex subset of a Banach space with the Radon–Nikodym property.

References

- [1] M. Martin and T. S. S. R. K. Rao, *On remotality for convex sets in Banach spaces*, Journal of Approximation Theory 162 (2010), 392–396; arXiv:0909.1992.

is nonremotal from 0 (alternatively, the set does not have any vector of maximal length). This gives an alternative proof of the main result from [7].

Remark 4. From the above arguments it is easy to see that for a weak*-compact set $E \subset X^*$ and for any $x^* \in X^*$, if the set $F(x^*, K)$ of farthest points in the weak*-closed convex hull K of E to x^* is non-empty, then it has a point of E . However the method of proof in [7] has the advantage that it shows that there is an extreme point of K in $F(x^*, K)$. Then by Milman's theorem [3, p. 151], such an extreme point is also in E .

The following easy example shows that the hypothesis of weak*-closedness can not be omitted on the set E in Theorem 1 (weak-compactness in the case of Remark 3).

Example 5. Let $\{e_n\}_{n \geq 1}$ denote the canonical vector basis in ℓ^2 . Let $X = \mathbb{K} \oplus_\infty \ell^2$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$ is the base field and $\oplus_\infty$ means the ℓ^∞ -direct sum. Consider the set

$$E = \left\{ \left(\frac{n}{n+1}, \frac{n}{n+1} e_n \right) : n \in \mathbb{N} \right\}.$$

Then E is a norm closed set which is not remotal from 0. Since $\overline{\text{co}}(E) = \overline{\text{co}}^{\text{weak}}(E)$ by Mazur's theorem and $\{e_n\}_{n \geq 1} \rightarrow 0$ in the weak topology, $(1, 0) \in \overline{\text{co}}(E)$ and so, $\overline{\text{co}}(E)$ is remotal from 0.

Remark 6. Let X be a Banach space and let $E \subset X$ be a weakly closed and bounded set. We do not know if remotality of $K = \overline{\text{co}}(E)$ from a point always implies that of E . Since any strongly exposed point of K clearly lies in E , the answer is affirmative if the farthest point in K is actually strongly exposed. We may also ask whether the above question has a positive answer for RNP sets (see [1, § 3] for these concepts).

We are now able to present the main result of our paper.

Theorem 7. Let X be an infinite dimensional Banach space. Then, there exists a closed and bounded convex set K that is not remotal.

Proof. As before, we will construct a closed and bounded set E which is not remotal from 0 and show that $K = \overline{\text{co}}(E)$ is also not remotal from 0.

In view of Remark 3 (or of [7]), we may assume without loss of generality that X has the Schur property. Since X is infinite dimensional, by Rosenthal's ℓ^1 Theorem (see [3, § XI]), X contains an isomorphic copy of ℓ^1 . Let $\|\cdot\|$ denote the norm on X . Now we will be done if we can construct in every Banach space $Y = (\ell^1, \|\cdot\|)$ isomorphic to ℓ^1 , a closed convex bounded set $K \subseteq Y$ which is not remotal from 0. Let us write τ for the weak*-topology of ℓ^1 as dual of c_0 inherited in Y . This is a locally convex topology on Y weaker than the norm topology and any τ -closed norm-bounded set is compact in this topology. Observe now that $\|\cdot\|$ is not necessarily weak*-lower semi-continuous (i.e. Y may not be a dual space) so, on the one hand, Corollary 2 does not apply and, on the other hand, B_Y may not be τ -closed.

Let $\{e_n\}_{n \geq 1}$ be the canonical basis of ℓ^1 . Consider the set

$$E = \left\{ \frac{n}{n+1} \frac{e_n}{\|e_n\|} : n \in \mathbb{N} \right\} \cup \{0\} \subseteq B_Y$$

Figure 1: The source page containing Remark 8 of arXiv:0909.1992.